
Conn2Conn Findings Ledger v2: A Section-by-Section F1–F10 Preprint

Adel Sahuc
Conn2Conn Preprint Draft v2

Abstract

This version rewrites the Conn2Conn preprint as a finding-by-finding ledger. Each section tracks one finding, F1 through F10, across the two parcellations where confirmatory data exist and across the model families or objective variants that bear on the claim. The main distinction is between confirmatory findings replicated on both Glasser and 4S456Parcels, and exploratory checks that are still Glasser-only or hypothesis-generating. The central conclusion is unchanged: cross-modal connectome prediction is real and directional, but reconstruction quality does not transfer into cognition utility. FC is the useful cognitive modality in this HCP-YA regime; structural connectomes, richer tractography, nonlinear models, and imputed connectomes do not clear the bv+demo floor.

1 Reader’s Map

This draft is intentionally longer than the compact preprint. It is a ledger rather than a teaser: every finding gets its own section, its own figure, a parcellation/model table where the repository contains such data, and an explicit status label.

- F1–F5 come from the confirmatory reproduction spine: two parcellations, ten frozen family-aware splits, and three model families where applicable.
- F6–F7 come from the family-mechanism grid: both parcellations, objective variants, permutation tests, and bootstrap intervals.
- F8 is intentionally hedged. The mechanism exists in both parcellations in the property sense, but the PC index migrates across atlases, so the claim is a mechanism hypothesis rather than a settled biomarker.
- F9 and F10 remain Glasser/exploratory sanity checks. They are still important because they close concrete escape routes: richer tractography, nonlinear model class, residual learning, cross-modal interactions, and sample-size scaling.

Table 1: F1-F10 status ledger.

Finding	Meaning	Status	Primary evidence
F1	FC->SC > SC->FC asymmetry	replicated both	reconstruction grid
F2	anatomy->SC / demographics->FC double dissociation	replicated both	reconstruction grid
F3	imputed connectomes are usable but utility is asymmetric	replicated both	downstream grid
F4	observed FC carries cognition signal	replicated both	downstream grid
F5	SC underperforms; imputation does not transfer cognition utility	replicated both	downstream grid
F6	predicted connectomes carry heritable family signal	replicated both	family grid
F7	reconstruction objective and identification objective trade off	replicated both	family grid
F8	exploratory PC mechanism, PC3 Glasser / PC4 4S456	partial	family/mechanism grid
F9	richer tractography representation does not help	Glasser/exploratory	tractography sanity
F10	nonlinear models and more data do not break ceiling	Glasser/exploratory	nonlinear sanity

2 F1: FC->SC Exceeds SC->FC

Status: replicated on both parcellations. The directional asymmetry is the scaffolding result: FC predicts individual SC deviations better than SC predicts individual FC deviations. The compact summary quotes the PCA-PLS workhorse because it is the original reconstruction model; v2 shows the model ledger explicitly. The direction survives PCA-PLS, Bayesian ridge, and the kernel-ridge family. The important point is not that the absolute value is identical across estimators. It is not. The important point is that every reasonable estimator family keeps the same ordering.

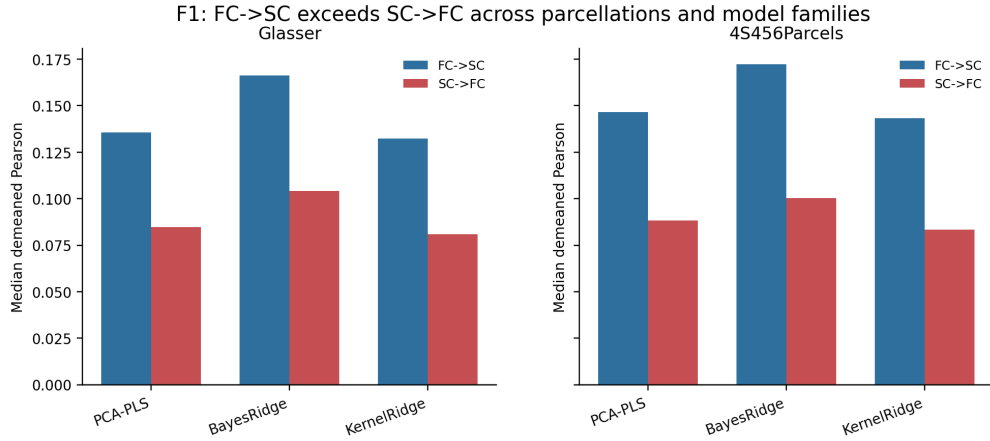


Figure 1: F1 across parcellations and model families. Bars show median demeaned Pearson for FC->SC and SC->FC. The FC->SC bar is higher in both atlases and across PCA-PLS, Bayesian ridge, and the kernel-ridge sweep.

Table 2: F1 median reconstruction metrics by parcellation and model family.

parcellation	model	FC->SC	SC->FC	ratio_FC_to_SC_over_SC_to_FC
4S456Parcels	BayesRidge	0.172	0.100	1.715
4S456Parcels	KernelRidge	0.143	0.083	1.723
4S456Parcels	PCA-PLS	0.146	0.088	1.660
Glasser	BayesRidge	0.166	0.104	1.596
Glasser	KernelRidge	0.132	0.081	1.633
Glasser	PCA-PLS	0.135	0.085	1.600

The 4S456 parcellation gives slightly higher structure-target prediction and the Glasser atlas gives nearly the same asymmetry ratio. This matters because the atlas comparison is within-cohort and split-matched. The finer atlas does not invent the effect; it shifts the absolute structural resolution while preserving the direction. Sources: reproduction/outputs/reconstruction.csv; reproduction/reports/reproduction_findings.md; reproduction/exploration/FINDINGS_EXPLORATION.md; notebooks-FC_to_SC-experimental/sanity_checks/preprocessing_check/reduction_axis_summary.csv; notebooks-FC_to_SC-experimental/sanity_checks/preprocessing_check/findings.md.

3 F2: Anatomy Predicts SC, Demographics Predict FC

Status: replicated on both parcellations. F2 is a crossed dissociation, not a single nuisance observation. Brain-volume/anatomy variables better predict structural connectivity, while demographics better predict functional connectivity. The clean way to write it is as a 2x2: anatomy has its natural target in SC; demographics have their natural target in FC. The same family-aware split logic and model families are used.

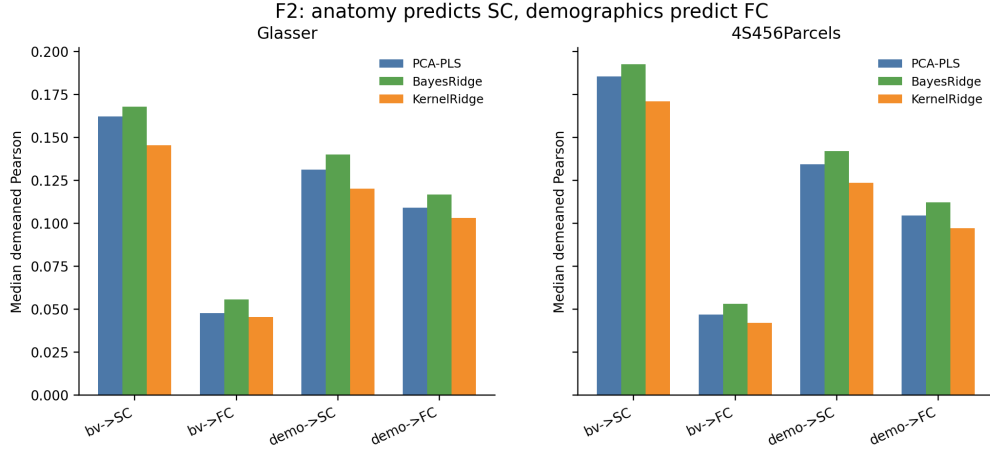


Figure 2: F2 across parcellations and model families. Anatomy/brain-volume predicts SC more strongly than FC, while demographics shift toward FC. The crossover is visible in both atlases.

Table 3: F2 model ledger: source variables to structural or functional targets. Showing first 18 rows.

parcellation	model	direction	demeaned_pearson	avg_rank	top1_acc
4S456Parcels	BayesRidge	bv->FC	0.053	0.658	0.015
4S456Parcels	BayesRidge	bv->SC	0.193	0.902	0.108
4S456Parcels	BayesRidge	demo->FC	0.112	0.673	0.015
4S456Parcels	BayesRidge	demo->SC	0.142	0.784	0.026
4S456Parcels	KernelRidge	bv->FC	0.042	0.612	0.013
4S456Parcels	KernelRidge	bv->SC	0.171	0.887	0.103
4S456Parcels	KernelRidge	demo->FC	0.097	0.666	0.015
4S456Parcels	KernelRidge	demo->SC	0.124	0.738	0.021
4S456Parcels	PCA-PLS	bv->FC	0.047	0.626	0.010
4S456Parcels	PCA-PLS	bv->SC	0.185	0.904	0.128
4S456Parcels	PCA-PLS	demo->FC	0.105	0.673	0.013
4S456Parcels	PCA-PLS	demo->SC	0.134	0.769	0.026
Glasser	BayesRidge	bv->FC	0.055	0.651	0.013
Glasser	BayesRidge	bv->SC	0.168	0.867	0.077
Glasser	BayesRidge	demo->FC	0.117	0.664	0.013
Glasser	BayesRidge	demo->SC	0.140	0.767	0.020
Glasser	KernelRidge	bv->FC	0.045	0.618	0.010
Glasser	KernelRidge	bv->SC	0.146	0.843	0.069

The dissociation is central because it turns bv+demo from a nuisance into a required scientific baseline. A connectome that cannot beat cheap subject-level co-variates should not be advertised as a useful cognitive biomarker. Sources: reproduction/outputs/reconstruction.csv; reproduction/exploration/FINDINGS_EXPLORATION.md; notebooks-FC_to_SC-experimental/MASTER_FINDINGS.md.

4 F3: Imputed Connectomes Are Usable, But Utility Is Asymmetric

Status: replicated on both parcellations, interpreted through F5. F3 is the bridge between reconstruction and downstream utility. Predicted connectomes are not random. They carry source-modality information and can support some downstream tasks. But the utility is not symmetric: imputed SC from FC is closer to neutral or mildly useful, whereas imputed FC from SC is often harmful for cognition. This is why the paper should not stop at reconstruction metrics.

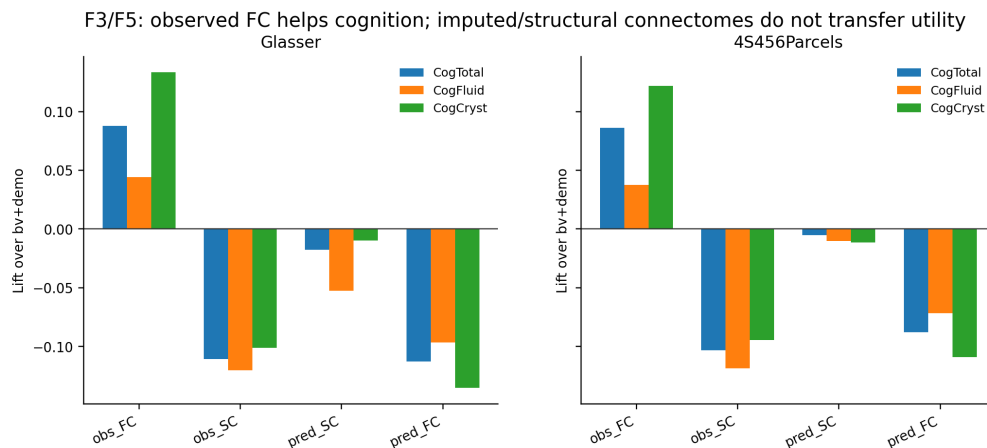


Figure 3: F3/F5 downstream utility ledger. Bayesian-ridge cognition lift over bv+demo is shown for observed and predicted FC/SC. Observed FC carries the positive signal; structural and imputed representations do not transfer it reliably.

Table 4: F3 downstream utility of observed and imputed connectomes, Bayesian ridge. Showing first 18 rows.

parcellation	input_set	target	pearson	lift_over_bvdemo	median_perm_p	residualized_pearson
4S456Parcels	obs_FC	CogCryst	0.476	0.122	0.041	0.352
4S456Parcels	obs_SC	CogCryst	0.259	-0.095	0.883	0.066
4S456Parcels	pred_FC	CogCryst	0.245	-0.109	0.896	0.025
4S456Parcels	pred_SC	CogCryst	0.342	-0.012	0.629	0.184
4S456Parcels	obs_FC	CogFluid	0.321	0.037	0.347	0.204
4S456Parcels	obs_SC	CogFluid	0.165	-0.118	0.971	0.030
4S456Parcels	pred_FC	CogFluid	0.211	-0.072	0.906	0.091
4S456Parcels	pred_SC	CogFluid	0.273	-0.010	0.843	0.152
4S456Parcels	obs_FC	CogTotal	0.445	0.086	0.090	0.289
4S456Parcels	obs_SC	CogTotal	0.256	-0.103	0.927	0.057
4S456Parcels	pred_FC	CogTotal	0.271	-0.088	0.871	0.079
4S456Parcels	pred_SC	CogTotal	0.354	-0.005	0.612	0.182
Glasser	obs_FC	CogCryst	0.487	0.133	0.031	0.378
Glasser	obs_SC	CogCryst	0.253	-0.101	0.817	0.061
Glasser	pred_FC	CogCryst	0.219	-0.135	0.960	-7.69e-05
Glasser	pred_SC	CogCryst	0.344	-0.010	0.582	0.195
Glasser	obs_FC	CogFluid	0.327	0.044	0.326	0.216
Glasser	obs_SC	CogFluid	0.163	-0.120	0.977	0.008

The model caveat is important. The scalar downstream grid includes PCA-PLS and kernel ridge rows, but the project ledger treats Bayesian ridge as the reportable scalar estimator because the discrepancy audit showed it is stable for cognition targets. The other model families are still useful as sensitivity checks; they do not rescue the imputation story. Sources: reproduction/outputs/downstream.csv; reproduction/exploration/FINDINGS_EXPLORATION.md; reproduction/reports/reproduction_findings.md.

5 F4: Observed FC Carries Real Cognition Signal

Status: replicated on both parcellations. F4 is the positive control for the entire cognition argument. If the pipeline could not detect any connectome-cognition signal at all, the SC/imputation null would be much less meaningful. Observed FC does clear the bv+demo floor, especially for crystallized cognition. This gives the negative claims teeth: the downstream framework can find signal when signal exists.

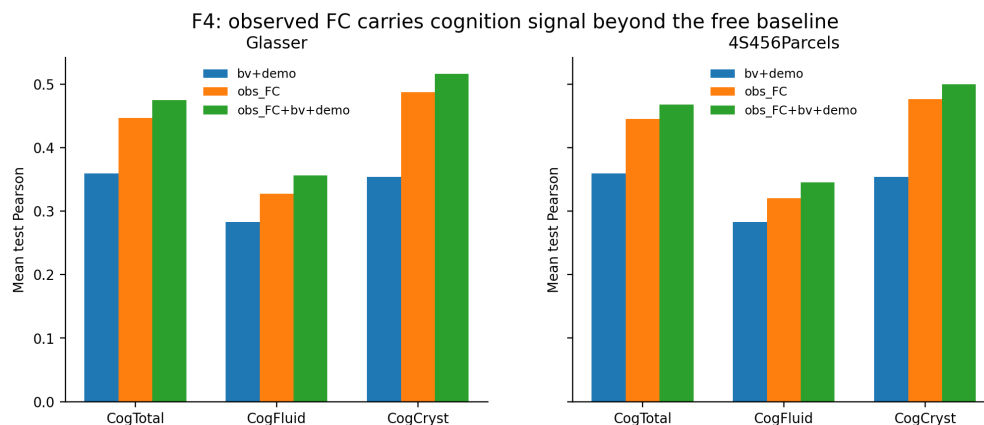


Figure 4: F4 observed FC cognition signal. Mean Bayesian-ridge Pearson is shown for bv+demo, observed FC, and observed FC plus bv+demo across cognition targets and parcellations.

Table 5: F4 observed FC vs bv+demo cognition ledger.

parcellation	input_set	target	pearson	lift_over_bvdemo	median_perm_p	residualized_pearson
4S456Parcels	bv+demo	CogCryst	0.354	0.000	1.000	0.054
4S456Parcels	obs_FC	CogCryst	0.476	0.122	0.041	0.352
4S456Parcels	obs_FC+bv+demo	CogCryst	0.500	0.146	0.008	0.353
4S456Parcels	bv+demo	CogFluid	0.283	0.000	1.000	-0.003
4S456Parcels	obs_FC	CogFluid	0.321	0.037	0.347	0.204
4S456Parcels	obs_FC+bv+demo	CogFluid	0.345	0.062	0.158	0.208
4S456Parcels	bv+demo	CogTotal	0.359	0.000	1.000	0.022
4S456Parcels	obs_FC	CogTotal	0.445	0.086	0.090	0.289
4S456Parcels	obs_FC+bv+demo	CogTotal	0.468	0.109	0.019	0.292
Glasser	bv+demo	CogCryst	0.354	0.000	1.000	0.054
Glasser	obs_FC	CogCryst	0.487	0.133	0.031	0.378
Glasser	obs_FC+bv+demo	CogCryst	0.516	0.162	0.002	0.382
Glasser	bv+demo	CogFluid	0.283	0.000	1.000	-0.003
Glasser	obs_FC	CogFluid	0.327	0.044	0.326	0.216
Glasser	obs_FC+bv+demo	CogFluid	0.356	0.073	0.095	0.219
Glasser	bv+demo	CogTotal	0.359	0.000	1.000	0.022
Glasser	obs_FC	CogTotal	0.447	0.088	0.080	0.297
Glasser	obs_FC+bv+demo	CogTotal	0.474	0.115	0.016	0.303

The result is intentionally modest. These are research-grade correlations, not a clinical biomarker. The point is comparative: FC is the modality that adds cognitive information above the cheap floor in this dataset; SC does not. Sources: reproduction/outputs/downstream.csv; notebooks-FC_to_SC-experimental/MASTER_FINDINGS.md; reproduction/exploration/FINDINGS_EXPLORATION.md.

6 F5: SC Underperforms, Imputation Does Not Transfer Cognition Utility

Status: replicated on both parcellations. F5 is the paper's main redirect. Observed SC is below the bv+demo floor for cognition, predicted SC is at best small and target-dependent, and predicted FC is often actively harmful. This is stronger than "imputation is useless." It says reconstruction can manufacture a plausible connectome whose downstream cognitive content is misaligned with what real FC carries.

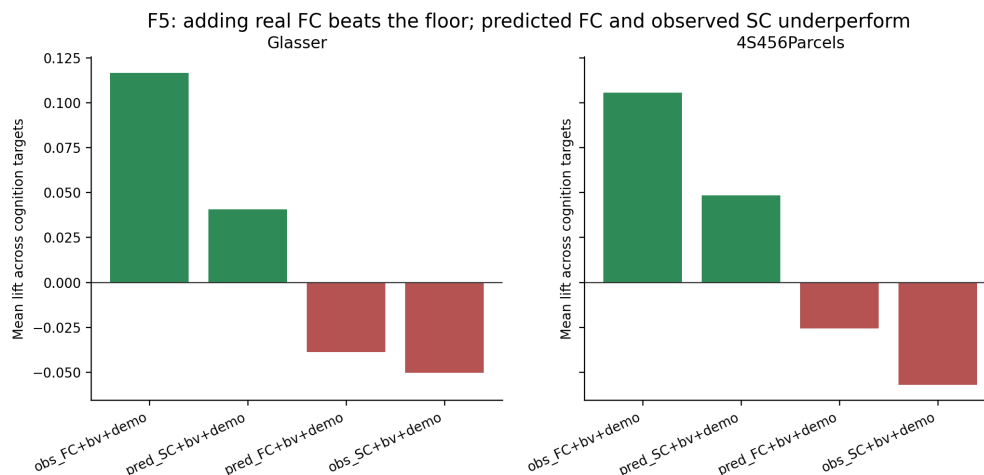


Figure 5: F5 cognition lift when each connectome is added to bv+demo. Observed FC gives a positive lift; predicted SC is smaller; predicted FC and observed SC underperform.

Table 6: F5 connectome plus bv+demo cognition ledger, Bayesian ridge. Showing first 18 rows.

parcellation	input_set	target	pearson	lift_over_bvdemo	median_perm_p
4S456Parcels	obs_FC+bv+demo	CogCryst	0.500	0.146	0.008
4S456Parcels	obs_SC+bv+demo	CogCryst	0.312	-0.042	0.695
4S456Parcels	pred_FC+bv+demo	CogCryst	0.326	-0.028	0.542
4S456Parcels	pred_SC+bv+demo	CogCryst	0.410	0.056	0.048
4S456Parcels	obs_FC+bv+demo	CogFluid	0.345	0.062	0.158
4S456Parcels	obs_SC+bv+demo	CogFluid	0.206	-0.077	0.934
4S456Parcels	pred_FC+bv+demo	CogFluid	0.260	-0.024	0.765
4S456Parcels	pred_SC+bv+demo	CogFluid	0.325	0.042	0.216
4S456Parcels	obs_FC+bv+demo	CogTotal	0.468	0.109	0.019
4S456Parcels	obs_SC+bv+demo	CogTotal	0.308	-0.052	0.831
4S456Parcels	pred_FC+bv+demo	CogTotal	0.334	-0.025	0.735
4S456Parcels	pred_SC+bv+demo	CogTotal	0.407	0.048	0.143
Glasser	obs_FC+bv+demo	CogCryst	0.516	0.162	0.002
Glasser	obs_SC+bv+demo	CogCryst	0.319	-0.035	0.604
Glasser	pred_FC+bv+demo	CogCryst	0.314	-0.040	0.813
Glasser	pred_SC+bv+demo	CogCryst	0.415	0.061	0.037
Glasser	obs_FC+bv+demo	CogFluid	0.356	0.073	0.095
Glasser	obs_SC+bv+demo	CogFluid	0.217	-0.066	0.937

This section should be written as an objective mismatch, not a model failure. F6 and F7 show why: predicted connectomes can carry real biological/family signal, but the signal selected by reconstruction is not the one needed for cognition. Sources: reproduction/outputs/downstream.csv; reproduction/exploration/FINDINGS_EXPLORATION.md; reproduction/outputs/leak_verdict.csv.

7 F6: Predicted Connectomes Carry Heritable Family Signal

Status: replicated on both parcellations. F6 prevents an over-simple reading of F5. Predicted connectomes are not empty artifacts. When the analysis asks whether they preserve family structure, the answer is yes. The strongest sibling result is the predicted SC residualized against bv+demo, which separates siblings from unrelated pairs well above the bv+demo-only baseline in both parcellations.

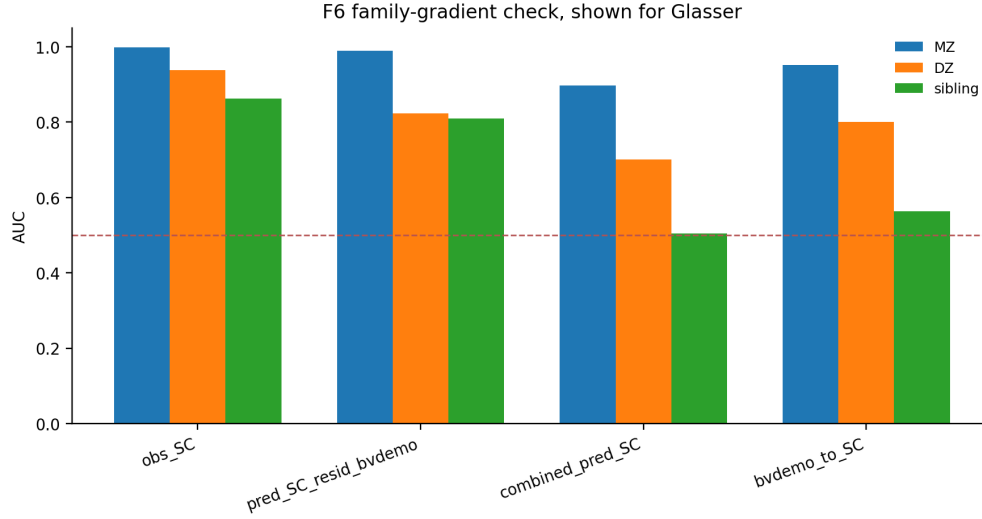


Figure 6: F6 family-gradient check. AUCs are shown for MZ, DZ, and sibling separation in selected variants. Predicted SC residualized against bv+demo preserves family signal beyond the subject-info baseline.

Table 7: F6 family AUCs with confidence intervals and FDR-adjusted permutation tests.

parcellation	variant	relation	auc	auc_lo	auc_hi	p_fdr	sig_fdr
4S456Parcels	bvdemo_to_SC	DZ	0.791	0.748	0.831	0.000	yes
4S456Parcels	bvdemo_to_SC	MZ	0.949	0.934	0.964	0.000	yes
4S456Parcels	bvdemo_to_SC	sibling	0.565	0.544	0.586	0.000	yes
4S456Parcels	obs_FC	DZ	0.886	0.851	0.916	0.000	yes
4S456Parcels	obs_FC	MZ	0.991	0.986	0.995	0.000	yes
4S456Parcels	obs_FC	sibling	0.821	0.805	0.836	0.000	yes
4S456Parcels	obs_SC	DZ	0.958	0.946	0.968	0.000	yes
4S456Parcels	obs_SC	MZ	0.999	0.999	1.000	0.000	yes
4S456Parcels	obs_SC	sibling	0.884	0.873	0.896	0.000	yes
4S456Parcels	pred_SC_resid_bvdemo	DZ	0.847	0.810	0.880	0.000	yes
4S456Parcels	pred_SC_resid_bvdemo	MZ	0.994	0.991	0.997	0.000	yes
4S456Parcels	pred_SC_resid_bvdemo	sibling	0.816	0.801	0.831	0.000	yes
Glasser	bvdemo_to_SC	DZ	0.800	0.759	0.840	0.000	yes
Glasser	bvdemo_to_SC	MZ	0.951	0.935	0.964	0.000	yes
Glasser	bvdemo_to_SC	sibling	0.563	0.541	0.585	0.000	yes
Glasser	obs_FC	DZ	0.886	0.852	0.914	0.000	yes
Glasser	obs_FC	MZ	0.990	0.984	0.994	0.000	yes
Glasser	obs_FC	sibling	0.823	0.807	0.839	0.000	yes
Glasser	obs_SC	DZ	0.938	0.924	0.951	0.000	yes
Glasser	obs_SC	MZ	0.999	0.998	1.000	0.000	yes
Glasser	obs_SC	sibling	0.863	0.850	0.876	0.000	yes
Glasser	pred_SC_resid_bvdemo	DZ	0.823	0.784	0.860	0.000	yes
Glasser	pred_SC_resid_bvdemo	MZ	0.989	0.982	0.994	0.000	yes
Glasser	pred_SC_resid_bvdemo	sibling	0.810	0.795	0.825	0.000	yes

The family result is constructive: cross-modal prediction can preserve a heritable connectome signature. The lesson is narrower and more interesting than "prediction is bad." Prediction preserves the signal selected by its objective and representation. Sources: reproduction/family_mechanism/outputs/family_auc.csv; dev-notes/PROJECT-HANDOFF-2026-06-22.md; notebooks-FC_to_SC-experimental/MASTER_FINDINGS.md.

8 F7: Reconstruction and Identification Objectives Trade Off

Status: replicated on both parcellations. F7 is the objective warning. The combined reconstruction-oriented predicted SC looks like a better reconstruction object, but sibling identification collapses to chance. The residualized predicted SC keeps family signal, while the combined reconstruction

variant does not. Thus one cannot assume that optimizing reconstruction, identification, and cognition are aligned.

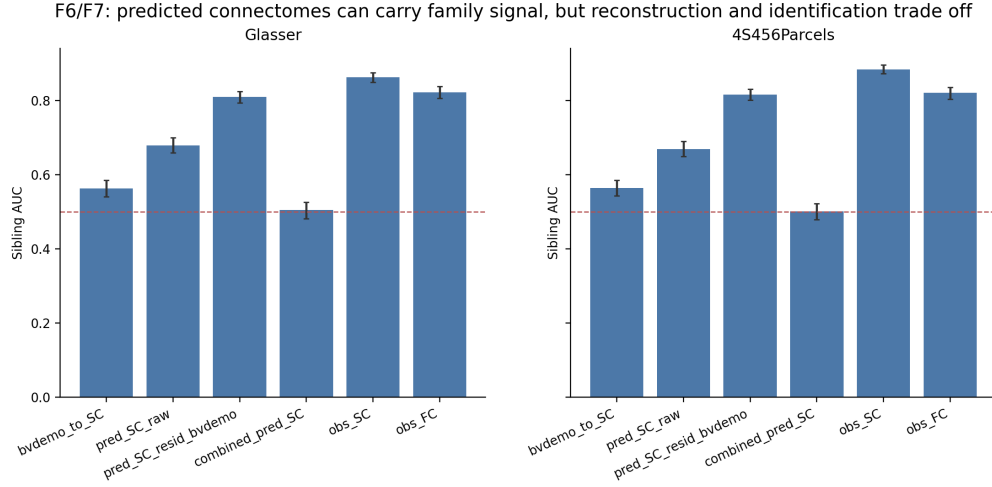


Figure 7: F6/F7 sibling AUC across objective variants. The combined predicted SC variant collapses to chance for sibling separation, while pred_SC_resid_bvdemo remains strongly above chance.

Table 8: F7 sibling AUC objective tradeoff.

parcellation	variant	auc	auc_lo	auc_hi	p_fdr	sig_fdr
4S456Parcels	bvdemo_to_SC	0.565	0.544	0.586	0.000	yes
4S456Parcels	combined_pred_SC	0.501	0.479	0.523	0.904	no
4S456Parcels	pred_SC_raw	0.670	0.650	0.690	0.000	yes
4S456Parcels	pred_SC_resid_bvdemo	0.816	0.801	0.831	0.000	yes
Glasser	bvdemo_to_SC	0.563	0.541	0.585	0.000	yes
Glasser	combined_pred_SC	0.505	0.482	0.526	0.648	no
Glasser	pred_SC_raw	0.680	0.660	0.700	0.000	yes
Glasser	pred_SC_resid_bvdemo	0.810	0.795	0.825	0.000	yes

This is the best conceptual support for the manuscript's redirect: the missing cognition transfer is not because the predicted connectomes are vacuous. It is because connectome objectives are not interchangeable. Sources: reproduction/family_mechanism/outputs/family_auc.csv; reproduction/exploration/FINDINGS_EXPLORATION.md; notebooks-FC_to_SC-experimental/MASTER_FINDINGS.md.

9 F8: Exploratory PC Mechanism, PC3 Glasser / PC4 4S456

Status: partial and hypothesis-generating. F8 should be written carefully. The mechanism-like structural PC exists in the property sense: a low-variance SC mode is moderately FC-predictable, carries weak family signal, and localizes to visual and dorsal attention edges. But the index is atlas-sensitive: the Glasser story is PC3, while the 4S456 mechanism shifts to a neighboring PC. Therefore the right claim is not "PC3 is the biomarker." The right claim is "there is a small, heritable, FC-predictable structural mode whose exact PC index depends on the atlas."

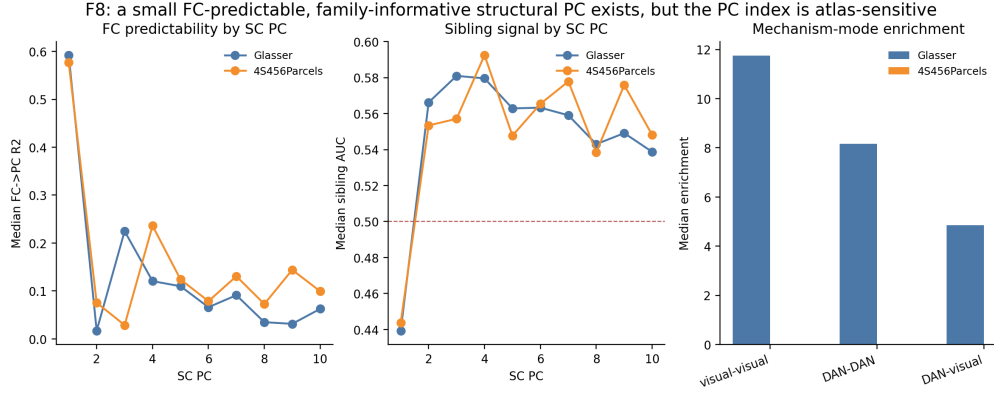


Figure 8: F8 mechanism ledger. FC predictability, sibling AUC, and network enrichment are shown across PCs and parcellations. The mechanism property replicates more clearly than the literal PC index.

Table 9: F8 PC stability, explained variance, FC predictability, and sibling AUC.

parcellation	anchor_pc	median_abs_cos	median_expl_var	median_FC_to_PC_R2	median_AUC_sibling
4S456Parcels	1	0.983	0.038	0.577	0.444
4S456Parcels	2	0.966	0.019	0.075	0.553
4S456Parcels	3	0.942	0.015	0.029	0.557
4S456Parcels	4	0.914	0.014	0.236	0.592
4S456Parcels	5	0.858	0.012	0.124	0.548
4S456Parcels	6	0.845	0.011	0.079	0.565
4S456Parcels	7	0.843	0.010	0.131	0.578
4S456Parcels	8	0.766	0.009	0.073	0.538
4S456Parcels	9	0.677	0.009	0.144	0.576
4S456Parcels	10	0.451	0.009	0.099	0.548
Glasser	1	0.985	0.038	0.592	0.439
Glasser	2	0.934	0.016	0.017	0.566
Glasser	3	0.886	0.015	0.225	0.581
Glasser	4	0.896	0.013	0.120	0.580
Glasser	5	0.868	0.012	0.110	0.563
Glasser	6	0.784	0.011	0.066	0.563
Glasser	7	0.712	0.010	0.091	0.559
Glasser	8	0.756	0.009	0.035	0.543
Glasser	9	0.628	0.009	0.031	0.549
Glasser	10	0.566	0.009	0.063	0.539

Table 10: F8 enrichment of the mechanism mode in visual and dorsal-attention network pairs.

parcellation	net_pair	median_enrichment	min_enrichment	median_n_obs
Glasser	dorsal attention dorsal attention	8.163	4.583	28.500
Glasser	dorsal attention visual	4.861	4.114	39.000
Glasser	visual visual	11.741	9.709	52.000

The hedging is not weakness; it is the scientific point. The analysis already caught a single-seed false positive in an earlier PC label. V2 therefore treats F8 as a forward map for future work rather than a load-bearing claim. Sources: reproduction/family_mechanism/outputs/f8_per_pc.csv; reproduction/family_mechanism/outputs/f8_stability.csv; reproduction/family_mechanism/outputs/f8_pc3_enrichment_agg.csv; reproduction/family_mechanism/outputs/f8_pc3_localization.csv; notebooks-FC_to_SC-experimental/sanity_checks/tract_check/findings.md.

10 F9: Richer Tractography Representation Does Not Help

Status: Glasser/exploratory, but internally strong. F9 asks whether SC counts are simply too crude. The answer in the current data is no. The named-bundle r2t representation predicts FC worse than

raw streamline counts, adds no marginal signal on top of count-SC, and fails downstream cognition. This closes the most obvious "better structural features would rescue the story" escape route for the tested representation.

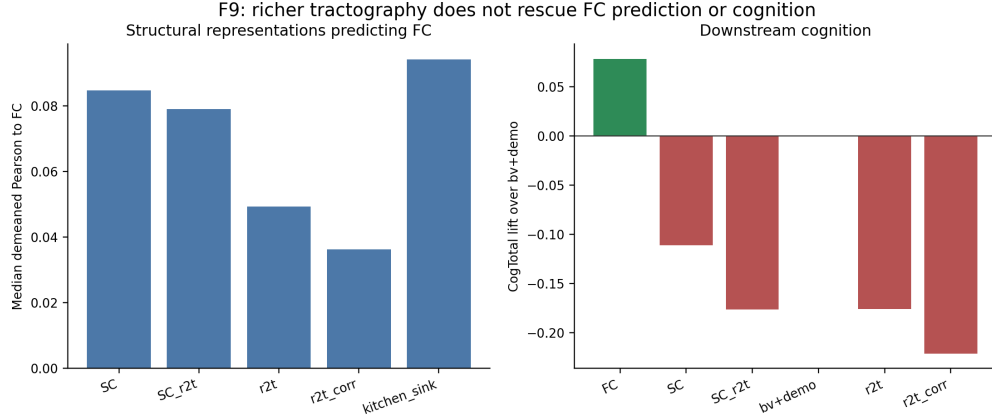


Figure 9: F9 tractography sanity check. Count-SC beats r2t bundle features for FC prediction, and all structural/tractography representations fall below the bv+demo cognition floor.

Table 11: F9 source representation prediction of FC, median across seeds.

rep	demeaned_pearson	pearson	top1_acc	avg_rank
kitchen_sink	0.094	0.829	0.049	0.723
SC	0.085	0.829	0.051	0.713
SC_r2t	0.079	0.827	0.036	0.702
r2t	0.049	0.815	0.021	0.609
r2t_corr	0.036	0.823	0.018	0.607

Table 12: F9 downstream cognition for tractography representations.

rep	target	pearson_raw	pearson_resid	lift_over_bvdemo_raw
FC	CogCrystalComp_Unadj	0.451	0.374	0.105
SC	CogCrystalComp_Unadj	0.267	0.086	-0.079
SC_r2t	CogCrystalComp_Unadj	0.160	0.028	-0.187
bv+demo	CogCrystalComp_Unadj	0.346		0.000
r2t	CogCrystalComp_Unadj	0.164	0.027	-0.183
r2t->synthFC	CogCrystalComp_Unadj	0.180		-0.166
r2t_corr	CogCrystalComp_Unadj	0.103	-0.009	-0.244
FC	CogFluidComp_Unadj	0.342	0.218	0.033
SC	CogFluidComp_Unadj	0.180	0.006	-0.130
SC_r2t	CogFluidComp_Unadj	0.135	0.072	-0.174
bv+demo	CogFluidComp_Unadj	0.309		0.000
r2t	CogFluidComp_Unadj	0.130	0.061	-0.179
r2t->synthFC	CogFluidComp_Unadj	0.197		-0.113
r2t_corr	CogFluidComp_Unadj	0.157	0.079	-0.152
FC	CogTotalComp_Unadj	0.451	0.264	0.078
SC	CogTotalComp_Unadj	0.261	0.040	-0.111
SC_r2t	CogTotalComp_Unadj	0.196	0.048	-0.177
bv+demo	CogTotalComp_Unadj	0.372		0.000
r2t	CogTotalComp_Unadj	0.196	0.049	-0.176
r2t->synthFC	CogTotalComp_Unadj	0.202		-0.170
r2t_corr	CogTotalComp_Unadj	0.151	-0.029	-0.221

Because F9 is Glasser-only, it should not be overclaimed as a universal tractography result. But it is exactly the right sanity check for this preprint: richer structural features do not recover FC's cognition signal in the tested regime. Sources: notebooks-FC_to_SC-experimental/tractography_predict/e1_source_rep_results.csv; notebooks-FC_to_SC-experimental/tractography_predict/e2_asymmetry_summary.csv; notebooks-FC_to_SC-experimental/tractography_predict/e3_marginal_summary.csv; notebooks-FC_to_SC-

11 F10: Nonlinear Models and More Data Do Not Break the Ceiling

Status: Glasser/exploratory, but broad within that scope. F10 closes the "bigger model" and "more data in this regime" escape routes. Kernel ridge preserves FC's known cognition signal but does not unlock tractography. Residual boosting gives the nonlinear model the linear answer for free and still finds no cognition residual. Cross-modal sink models dilute FC rather than improve it. The sample-size curve shows the nonlinear gap does not grow into positive territory through the tested n.

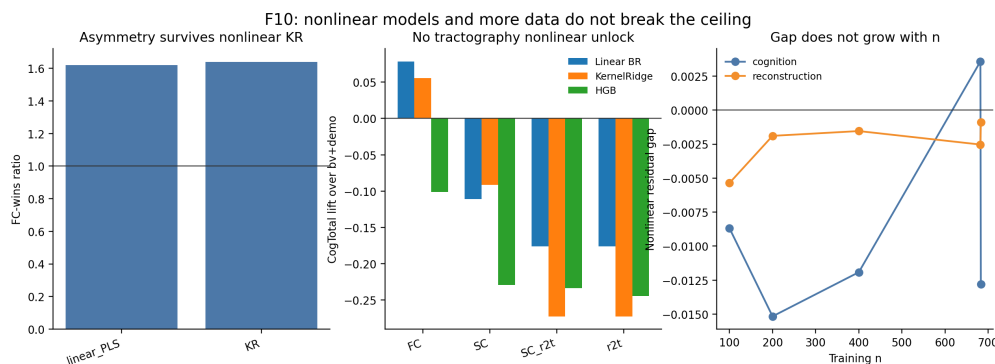


Figure 10: F10 nonlinear and scaling checks. KernelRidge preserves the asymmetry but does not unlock tractography cognition. The residual nonlinear gap stays at or below zero as n increases.

Table 13: F10 residual-boost reconstruction improvements. The small positive reconstruction sliver is far below the practical threshold.

rep	direction	median_template	median_final	median_improvement	wilcoxon_p_improve
SC	FC->SC	0.135	0.141	0.005	9.77e-04
SC	SC->FC	0.085	0.087	0.002	9.77e-04
r2t	FC->r2t	0.111	0.117	0.006	9.77e-04
r2t	r2t->FC	0.049	0.053	0.004	9.77e-04
r2t_corr	FC->r2t_corr	0.058	0.062	0.005	9.77e-04
r2t_corr	r2t_corr->FC	0.036	0.040	0.004	9.77e-04

Table 14: F10 sample-size scaling summary.

task	n_sub	n_seeds	median_linear	median_final	median_gap	wilcoxon_p_gap_gt0
cognition	100	10	0.302	0.290	-0.009	1.000
cognition	200	10	0.327	0.324	-0.015	0.993
cognition	400	10	0.394	0.380	-0.012	0.999
cognition	682	1	0.482	0.486	0.004	0.500
cognition	683	9	0.391	0.375	-0.013	0.990
reconstruction	100	10	0.041	0.035	-0.005	0.999
reconstruction	200	10	0.067	0.065	-0.002	0.884
reconstruction	400	10	0.094	0.093	-0.002	0.990
reconstruction	682	1	0.110	0.107	-0.003	1.000
reconstruction	683	9	0.110	0.110	-8.87e-04	0.980

F10 should be framed as a structural ceiling for this HCP-YA, normal-range cognition, cross-sectional setting. It does not prove there is no signal in clinical, longitudinal, or much larger cohorts. It does prove that the ordinary escape routes do not explain the current null. Sources: notebooks-FC_to_SC-experimental/non-linear-sanity-check/n1_cognition_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/n2_reconstruction_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/n3_marginal_summary.csv;

notebooks-FC_to_SC-experimental/non-linear-sanity-check/n4_cog_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/n4_recon_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/n5_cog_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/n5_recon_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/n6_scaling_summary.csv; notebooks-FC_to_SC-experimental/non-linear-sanity-check/findings_nonlinear.md; notebooks-FC_to_SC-experimental/non-linear-sanity-check/findings_residual.md; notebooks-FC_to_SC-experimental/non-linear-sanity-check/findings_scaling.md; notebooks-FC_to_SC-experimental/sanity_checks/noise_sanity_check/findings_noise.md.

12 Cross-Cutting Sanity Checks

Several sanity checks are not standalone findings but protect the ledger. The reduction-axis analysis shows F1 is not a PCA artifact. The FC reliability analysis shows the SC->FC gap is not explained by per-subject FC reliability. The leak verdict grid shows zero genuine leak failures.

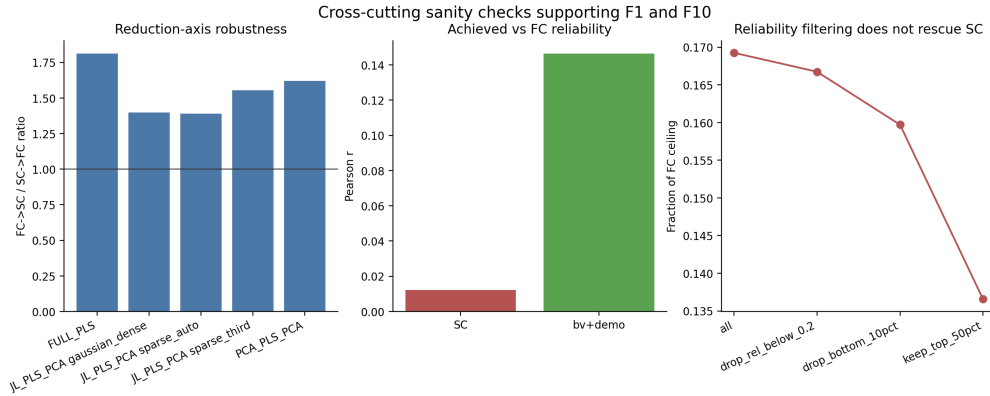


Figure 11: Cross-cutting sanity checks. Reduction choice preserves the F1 ratio; per-subject FC reliability does not explain SC->FC achieved performance; reliability filtering does not rescue SC.

Table 15: Reduction-axis robustness supporting F1.

method	jl_variant	n_seeds	median_ratio	min_ratio	max_ratio	wilcoxon_p_vs_1	median_FC_to_SC_dp	median_SC_to_FC_dp
FULL_PLS		10.000	1.813	1.493	2.115	9.77e-04	0.138	0.076
JL_PLS_PCA	gaussian_dense	10.000	1.397	1.194	2.250	9.77e-04	0.084	0.061
JL_PLS_PCA	sparse_auto	10.000	1.389	1.109	1.903	9.77e-04	0.084	0.059
JL_PLS_PCA	sparse_third	10.000	1.554	1.235	1.776	9.77e-04	0.085	0.053
PCA_PLS_PCA		10.000	1.621	1.324	1.861	9.77e-04	0.135	0.085

Table 16: Per-subject FC achieved-vs-ceiling correlations supporting F10.

source	n	pearson_achieved_vs_ceiling	pearson_p	spearman_achieved_vs_ceiling	spearman_p
SC	857	0.012	0.722	-9.67e-04	0.977
bv+demo	857	0.147	1.66e-05	0.147	1.66e-05

13 Limitations and Exact Next Experiments

- SC-side reliability remains data-blocked.** FC reliability is well measured, but a true SC test-retest or tractography perturbation ceiling is still needed.
- F9 and F10 should be ported to both parcellations.** The Glasser results are internally coherent, but the v2 ledger should label them exploratory until the same nonlinear/tractography checks run under 4S456Parcels.

- **F8 needs property-selected localization.** The mechanism mode should be selected by properties such as non-confounded FC predictability and family signal, not by hard-coded PC index.
- **Objective interpolation is the cleanest new experiment.** Train along a ladder from reconstruction to cognition to fingerprint/family objectives, then ask where the signals diverge.
- **External validity is not solved.** HCP-YA is healthy, cross-sectional, and normal-range cognition. The protocol should be repeated in clinical, longitudinal, or larger cohorts with the bv+demo baseline carried along.

14 Conclusion

The F1–F10 ledger supports a sharper, more disciplined manuscript than the compact draft. The data do not say that cross-modal prediction is impossible. They say that cross-modal prediction is directional, objectively real, and biologically structured, but that reconstruction is the wrong proxy for downstream cognition. Observed FC is the useful cognition modality in this regime. SC and richer structural representations are valuable for other biological questions, including family structure and mechanism hypotheses, but they do not transfer FC’s cognitive utility through imputation. The scientific redirect is therefore concrete: report bv+demo, stop treating reconstruction as utility, and move any future cognition claim to regimes where structural variation plausibly exceeds the healthy young-adult floor.